WAYGROUND formerly Quizizz **Worksheets**

Ionic Bonding

Total questions: 20

Worksheet time: 10mins

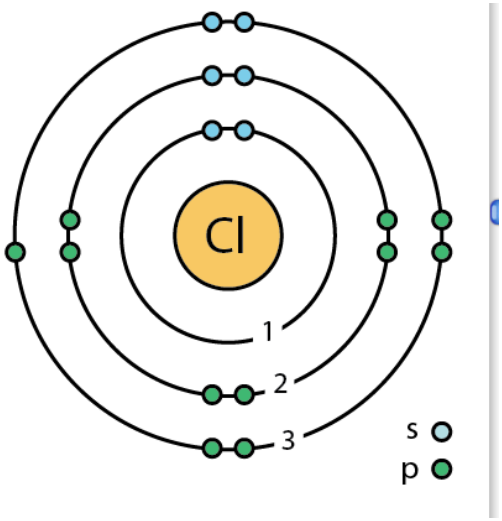
Instructor name: So Korm

Name

Class

Date

1.



How many **CORE ELECTRONS** does Chlorine have?

a) 4

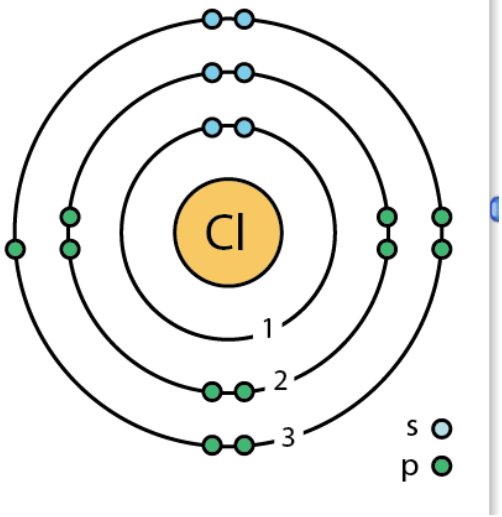
b) 10

c) 2

d) 8

e) 17

2.



How many **VALENCE ELECTRONS** does Chlorine have?

a) 8

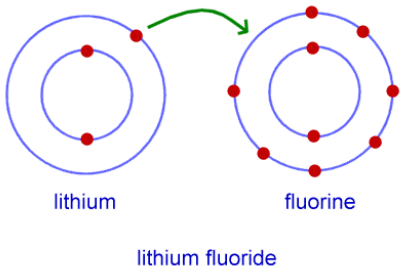
b) 10

c) 2

d) 17

e) 7

3.



Ionic bonds are formed when **electrons** get

a) **Transferred** from one atom to another

b) **Shared** between two atoms

c) **Added** to both atoms

d) **Removed** from both atoms

4. How many **valence electrons** does Mg have? (use the PTable)

a) 1

b) 3

c) 5

d) 4

e) 2

5. How many **valence electrons** does Po (Polonium) have? (Use your PTable)

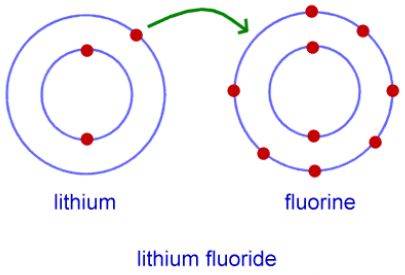
a) 6

b) 2

c) 4

d) 8

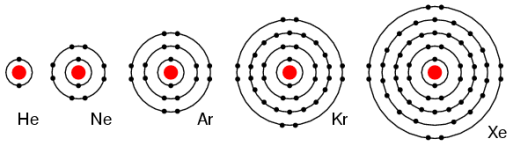
6.



Ionic bonds typically form between

a) Two nonmetals
b) A metalloid and a metal
c) Two metals
d) A metal and a nonmetal

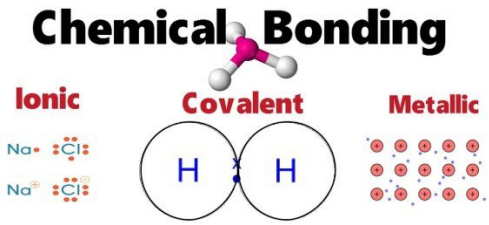
7.



Atoms are most stable when their outer shell is complete.

a) True
b) False

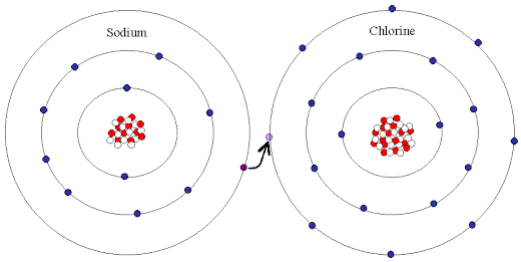
8.



Why do elements form chemical bonds?

a) To create a new element
b) Because they're friendly
c) Because they all need to gain more electrons
d) To become stable

9.



What is the correct formula for **Sodium Chloride**?

a) $NaCl_2$
b) SC
c) NaCl
d) SCl

10.

Name this ionic compound: Al_2O_3

a) Aluminum oxygen

b) Dialuminum oxide

c) Oxygen Aluminide

d) Aluminum oxide

11. If you were to draw the **lewis dot structure** for Germanium (Ge), how many dots would you put around it?

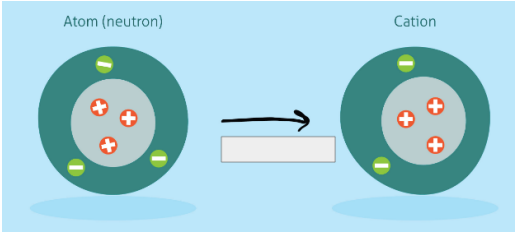
a) 6

b) 4

c) 8

d) 2

12.



Cations _____ electrons, becoming _____ charged.

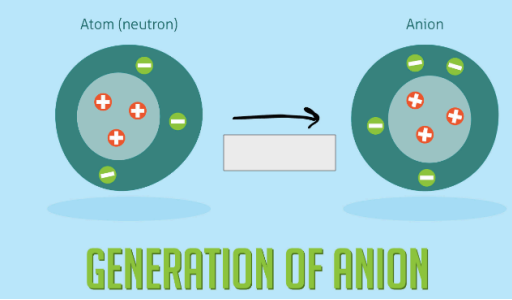
a) Gain; negatively (-)

b) Gain; positively (+)

c) Lose; negatively (-)

d) Lose; positively (+)

13.



Anions ____ electrons, becoming _____ charged.

a) Gain; negatively (-)

b) Lose; positively (+)

c) Gain; positively (+)

d) Lose; negatively (-)

14.

Write the chemical formula for K and Br (potassium bromide).

a) KBr_2

b) K_2Br

c) KBr_3

d) KBr

15.

Write the chemical formula for Na and S (sodium sulfide).

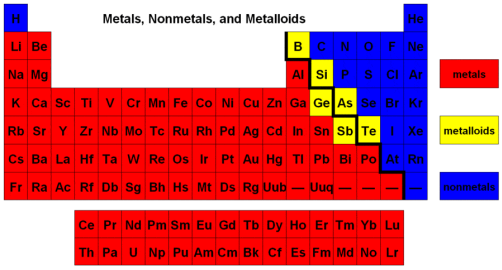
a) Na_2S

b) NaS_2

c) NaS

d) S_2S_2

16.



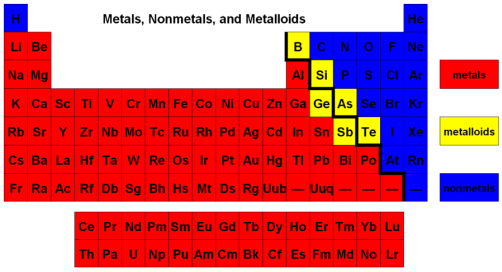
Metals (like Na) typically _____ electrons in the formation of ionic bonds.

a) Gain

b) Lose

c) Keep the same

17.



Nonmetals (like Cl) typically ____ electrons when they form ionic bonds.

a) Gain

b) Keep the same

c) Lose

18.

Why are there two sodiums (Na), in the formation of this ionic compound?

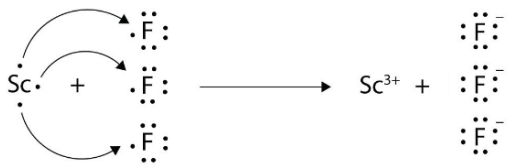
a) There always has to be 2 metals

b) The overall charge has to be 0

c) It needs to have more metal atoms than nonmetal

d) The overall charge needs to add up to 4

19.



Using the model for the formation of **Scandium Fluoride** (an ionic compound), write the **chemical formula**.

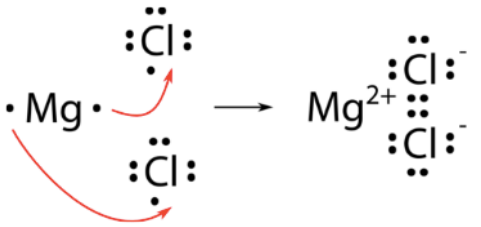
a) ScF_3

b) $Sc_{+3}F_{-1}$

c) Sc_3F_3

d) Sc_3F

20.



Using the model, write the **chemical formula** for an ionic compound that has Mg and Cl.

a) $Mg_{+2}Cl_{-1}$

b) Mg_2Cl_2

c) Mg_2Cl_2

d) $MgCl_2$

Answer Keys

1. b) 10
2. e) 7
3. a) **Transferred** from one atom to another
4. e) 2
5. a) 6
6. d) A metal and a nonmetal
7. a) True
8. d) To become stable
9. c) NaCl
10. d) Aluminum oxide
11. b) 4
12. d) Lose; positively (+)
13. a) Gain; negatively (-)
14. d) KBr
15. a) Na_2S
16. b) Lose
17. a) Gain
18. b) The overall charge has to be 0
19. a) ScF_3
20. d) $MgCl_2$